

Practicum 6: Current-Condition Operational Flags

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Question

What operational concerns are present in the latest saved weather observation?

Summary

The saved observation has no project-defined caution or severe flags. Its overall current-condition status is normal.

Data and method

The observation is a committed offline fixture originally obtained from the [Aviation Weather Center Data API](#). Rendering does not call the live API. The report loads the saved JSON and uses the project's existing `validate_metar_observation()` and `assess_current_conditions()` functions; it does not reproduce validation or assessment rules.

- **Saved file:** `tests/fixtures/metar-katl-success.json`
- **Station:** KATL (Atlanta/Hartsfield-Jackson Intl, GA, US)
- **Observation timestamp:** `2026-07-29T20:00:00.000Z`
- **Original request parameters:** `ids=KATL, format=json`
- **Units:** visibility in statute miles (SM), ceiling in feet above ground level (ft AGL), and wind in knots (kt)
- **Report render date:** `2026-08-02`

Current-condition evidence

Table 1: Observed conditions and project-generated operational statuses

Condition	Observed	Caution boundary	Result
Visibility	9 SM	Below 5 SM	Normal
Ceiling	20,000 ft AGL	Below 3,000 ft AGL	Normal
Sustained wind	9 kt	15 kt or more	Normal combined wind flag
Wind gust	Not reported	20 kt or more	Normal combined wind flag

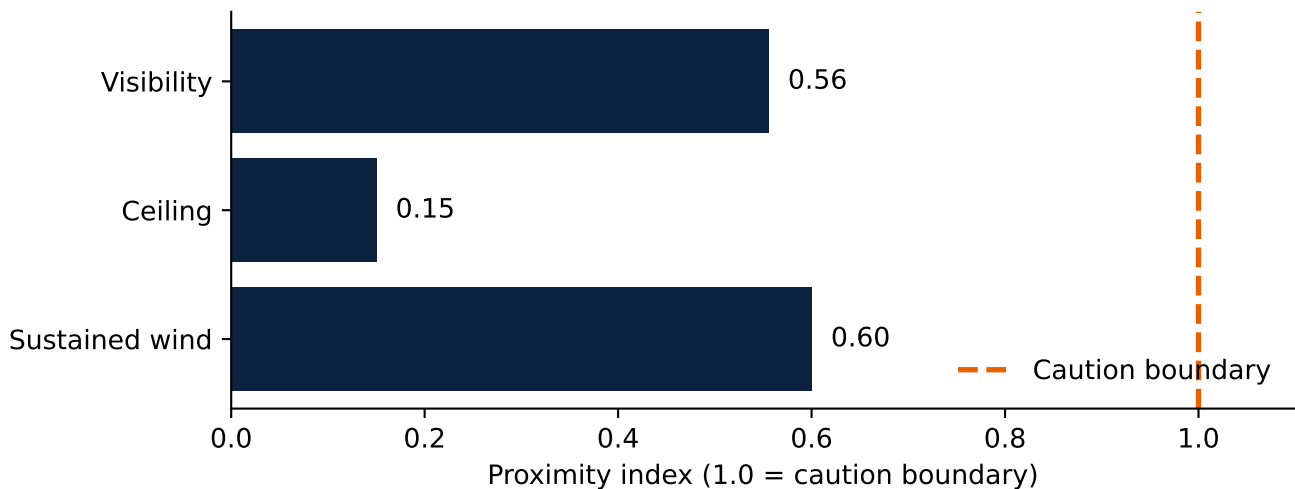


Figure 1: Observed values normalized to their project caution boundaries. The index is for visualization only; operational statuses come from the project assessment.

For visibility and ceiling, lower observations are more concerning, so the figure divides the caution boundary by the observation. For sustained wind, higher observations are more concerning, so it divides the observation by the caution boundary. A value of 1.0 reaches the caution boundary. Gust is omitted because no gust value was reported. This normalization supports visual comparison only and does not determine any flag.

Interpretation

KATL reported 9 SM visibility, a 20,000 ft AGL ceiling, and 9 kt sustained wind. Each assessable condition remains on the normal side of its project caution boundary, and the project assessment returns an overall status of **normal** with complete data. No gust was reported; the combined wind flag remains assessable from the sustained-wind value. These results are informational and are not official flight guidance.

Rebuild

After installing the locked dependencies with `uv sync`, rebuild the PDF from the repository root with:

```
uv run quarto render reports/practicum-6.qmd --to pdf --output-dir ../output/pdf
```